

Is Brain–Heart Coupling a Stable Physiological Trait?

Within-Night Variability, Between-Person Differences, and Long-Term Reproducibility Across Polysomnograms

Study protocol / registered-report manuscript

Abstract

Brain–heart coupling is usually compared between groups or sleep stages, although its most basic source of variation is unknown: does coupling fluctuate mainly within one person over a night, or does it differ mainly between people? We will divide artifact-screened polysomnography (PSG) into consecutive non-overlapping 5-minute windows and calculate a prespecified coupling metric in every usable window. A random-intercept model will partition total coupling variance into between-subject and within-subject components. The intraclass correlation coefficient (ICC) will quantify the fraction of total variance attributable to stable person-level differences. Stage-adjusted, stage-specific, and multivariate topographic analyses will distinguish an individual fingerprint from differences in sleep architecture. Repeat PSGs separated by months or years will test whether both coupling level and spatial fingerprint reappear. The primary estimand is the subject ICC with a bootstrap confidence interval; the longitudinal estimand is test–retest reliability across visits. This design turns an intuitive claim—that brain–heart coupling may be an individual physiological trait—into a falsifiable variance-components question.

Keywords: *brain–heart coupling; polysomnography; heartbeat-evoked potential; intraclass correlation; variance components; test–retest reliability; physiological fingerprint; sleep*

1. Introduction

The brain and heart interact continuously through autonomic, baroreceptive, respiratory, and central neural pathways.[1,2] During sleep, this interaction can be estimated under relatively standardized behavioral conditions and across several recurring physiological states.[3] Yet a basic measurement question remains unresolved. A coupling value observed in one segment may reflect a stable characteristic of the individual, transient variation over the night, sleep stage, recording noise, or cardiac field contamination of the electroencephalogram (EEG).[4,5]

Most studies summarize a recording at the participant level or test average differences between sleep stages and clinical groups. Averaging improves precision but hides the variation needed to determine whether coupling behaves like a trait. Sleep EEG spectra themselves show strong individual specificity across nights,[6–8] but it is unknown whether brain–heart coupling has comparable individuality. Repeated measurement within a night permits a direct decomposition: how much variability lies between people, and how much lies among windows from the same person? If between-person variance dominates after accounting for sleep state and technical covariates, coupling may be a physiological fingerprint. If within-person variance dominates, a single nightly summary is unlikely to characterize an individual reliably.

We propose a two-timescale analysis. First, coupling will be calculated every 5 minutes throughout each PSG. A hierarchical model will partition variance into between-subject and within-subject components and estimate the intraclass correlation coefficient (ICC). Second, repeat PSGs will test whether the inferred person-specific coupling level and multichannel pattern return after months or years. The second test is essential: a high within-night ICC can arise from stable properties of one recording session, including montage, electrode impedance, posture, or persistent artifact. Long-term re-identification across independently acquired nights provides a stronger trait criterion.

Two-timescale test of a brain-heart coupling fingerprint



$$\text{Within night: } y_{ij} = \beta_0 + u_i + \varepsilon_{ij}$$

$$ICC = \sigma_{\text{subject}}^2 / (\sigma_{\text{subject}}^2 + \sigma_{\text{within}}^2)$$

Long term: absolute agreement + fingerprint identification

Figure 1. Analysis overview. Every PSG is reduced to quality-controlled, non-overlapping 5-minute coupling estimates. A hierarchical model separates person-level from window-level variance; independent repeat PSGs then test whether the scalar level and multichannel fingerprint return.

2. Objectives and hypotheses

2.1 Primary objective

Quantify the proportion of total 5-minute brain–heart coupling variance attributable to differences between subjects.

2.2 Secondary objectives

1. Determine how much of the between-subject variance remains after adjustment for sleep stage, time of night, age, sex, cohort, respiratory burden, heart rate, and signal-quality measures.
2. Estimate ICC separately in wake, N1, N2, N3, and REM sleep.
3. Test whether person-level coupling estimates and multichannel coupling fingerprints reproduce in repeat PSGs separated by months or years.
4. Compare the observed fingerprint with negative controls designed to reveal cardiac field artifact, temporal autocorrelation, and recording-site effects.

2.3 Prespecified hypotheses

H1: Between-subject variance is greater than zero and the adjusted ICC is meaningfully above zero.

H2: The adjusted ICC is higher within a fixed sleep stage than in the unadjusted analysis because stage transitions contribute to within-person variability.

H3: Subject-level coupling values and spatial fingerprints show positive test–retest reliability across repeat PSGs.

Trait interpretation rule: We will describe coupling as supporting an individual physiological trait only if (a) the adjusted within-night ICC is substantial, (b) its confidence interval excludes values compatible with negligible reliability, and (c) repeat-night reliability is positive and robust to artifact controls. A high ICC alone will not be treated as sufficient evidence.

Illustration of the primary estimand (simulated data; not results)

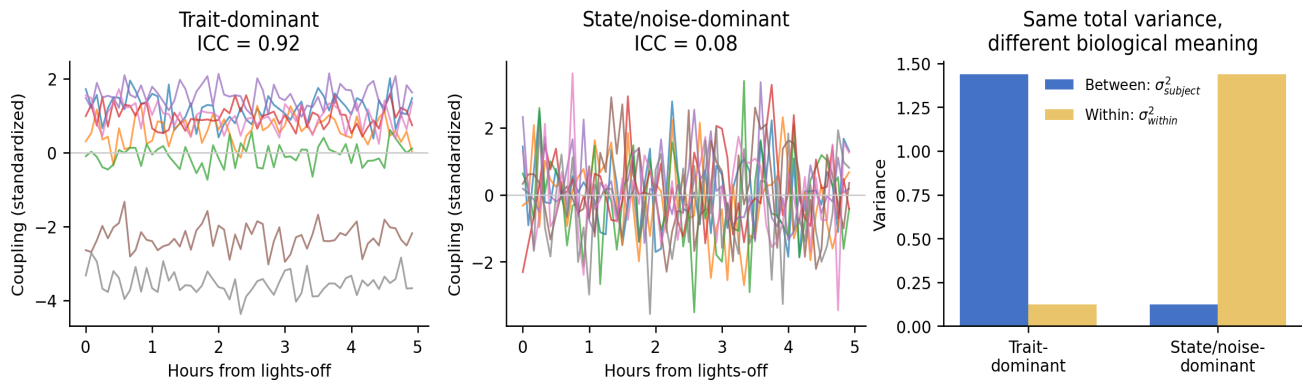


Figure 2. Meaning of the primary estimand. Simulated trajectories illustrate two datasets with similar total variance but opposite allocations to between-subject and within-subject components. Values are explanatory simulations, not study results.

3. Methods

3.1 Study population

We will include PSGs containing synchronized EEG and ECG, sleep-stage annotations, and at least six usable 5-minute windows after quality control. Large-scale harmonized sleep resources demonstrate the feasibility of such analyses across heterogeneous PSG cohorts.[9] The primary analysis will use one PSG per subject, selected using a prespecified rule that does not depend on coupling (prefer the earliest technically adequate diagnostic night). Repeat studies will be reserved for the longitudinal analysis. Studies with gross synchronization failure, absent ECG, unrecoverable channel metadata, or insufficient usable data will be excluded. Cohort-specific inclusion and exclusion counts will be reported in a flow diagram.

3.2 Five-minute windows

Each recording will be divided from lights-off into consecutive, non-overlapping 300-second windows. This duration balances temporal resolution against the number of heartbeats required for a stable estimate. Windows will not be shifted to align with stage boundaries. The primary stage label will be the modal 30-second sleep stage; stage purity, defined as the proportion of epochs in the modal stage, will be retained. The primary stage-specific analysis will require at least 80% purity. Sensitivity analyses will use 70%, 90%, and exclusively single-stage windows.

For every window we will retain clock time from lights-off, normalized night position, modal stage, stage purity, transition status, median heart rate, heart-rate variability, respiratory-event burden, oxygen saturation, arousal burden, channel-level quality metrics, and the number of accepted heartbeats.

3.3 Signal preprocessing

EEG and ECG preprocessing will be fixed before outcome modeling. EEG will be referenced consistently within each cohort, band-pass filtered using prespecified zero-phase filters, and resampled only when required to harmonize sampling rates. R peaks will be detected from ECG and subjected to refractory-period, morphology, and RR-interval checks. EEG segments containing clipping, flat signal, excessive high-frequency power, extreme amplitude, or large discontinuities will be rejected. Windows will be excluded if they contain fewer than a prespecified number of accepted beats or fail channel-quality thresholds.

Because ECG electrical activity can volume-conduct into scalp EEG, the primary neural coupling analysis will be performed after an ECG-informed cardiac-field-artifact correction.[4,5] Results will also be shown before correction. The correction parameters will be estimated without using subject identity or outcomes. A control analysis will quantify direct ECG-to-EEG waveform similarity; a purported trait that disappears after this control will be interpreted as a recording or anatomy-dependent cardiac artifact rather than neural coupling.

3.4 Primary coupling metric

The primary metric will be a heartbeat-evoked EEG coupling strength calculated independently in every 5-minute window. EEG will be epoched around accepted R peaks. For each channel, the post-heartbeat waveform in the prespecified neural interval will be compared with its pre-heartbeat baseline, and effect size will be expressed as baseline-normalized root-mean-square heartbeat-locked activity. Channel values will be combined using a prespecified robust average across the common scalp montage. The metric will be transformed if needed to obtain approximately symmetric model residuals. Heart-rate and heart-rate-variability features will follow established measurement standards.[10]

The exact filters, epoch boundaries, baseline interval, post-heartbeat interval, artifact thresholds, minimum heartbeat count, and channel set will be frozen in a machine-readable analysis configuration before outcome inspection. This prevents selecting a definition because it produces a favorable ICC.

Two complementary metrics will be treated as secondary: (1) ECG-to-EEG predictive coupling estimated by cross-validated explained variance, primarily as an artifact-sensitive comparator, and (2) information-theoretic or phase-based coupling between beat-to-beat cardiac dynamics and EEG amplitude, computed without cardiac time-locking. Agreement across metrics would strengthen generalizability; disagreement would clarify which physiological component is stable.

3.5 Variance decomposition

Let y_{ij} denote coupling for subject i in 5-minute window j . The unadjusted random-intercept model is

$$y_{ij} = \beta_0 + u_i + \epsilon_{ij},$$

where $u_i \sim N(0, \sigma_{\text{subject}}^2)$ is the subject-specific deviation and $\epsilon_{ij} \sim N(0, \sigma_{\text{within}}^2)$ is window-level variation. Thus,

$$\text{Var}(y) = \sigma_{\text{subject}}^2 + \sigma_{\text{within}}^2,$$

and the intraclass correlation coefficient is

$$\text{ICC} = \sigma_{\text{subject}}^2 / (\sigma_{\text{subject}}^2 + \sigma_{\text{within}}^2).$$

ICC is the expected correlation between two exchangeable 5-minute windows from the same subject under the model. ICC form, model, unit, and agreement definition will be reported explicitly because different ICC formulations answer different questions.[11–13] We will report both variance components, their ratio, ICC, and 95% confidence intervals. Confidence intervals will be obtained by subject-level parametric bootstrap, preserving the unequal number of windows per subject.

The adjusted model will include sleep stage, stage purity, normalized time of night and its nonlinear term, heart rate, respiratory-event burden, arousal burden, signal quality, cohort/site, age, and sex. Continuous covariates will be centered and modeled nonlinearly when justified in advance. Subject will remain a random intercept. Because adjacent windows can be serially correlated, the primary adjusted model will use an autoregressive within-night residual structure when supported by diagnostics; a block-bootstrap sensitivity analysis will resample subjects and contiguous runs of windows.

The unadjusted ICC answers how reproducible two randomly selected windows from the same night are in practice. The adjusted ICC asks whether individuals differ after known state and technical factors are held constant. Both are scientifically meaningful and will be reported without choosing between them post hoc.

State and artifact must not masquerade as identity

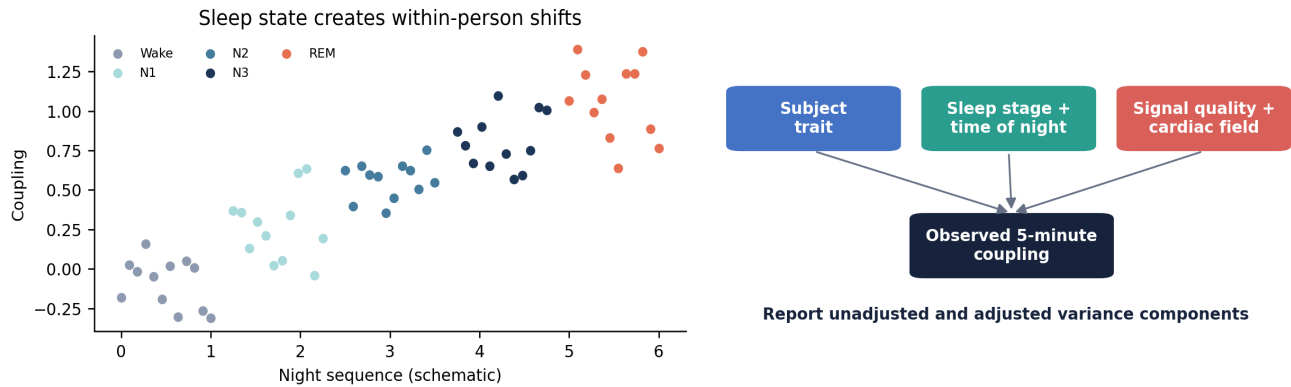


Figure 3. Adjustment strategy. Sleep stage and time of night produce genuine state shifts, whereas signal quality and cardiac field contamination can create technical stability. Unadjusted and adjusted variance components answer complementary questions. The left panel is schematic, not observed data.

3.6 Sleep-stage and temporal analyses

Separate variance components will be estimated for wake, N1, N2, N3, and REM among subjects with sufficient data in each stage. A joint model will test subject-by-stage random effects, asking whether a person who has high coupling in one stage also tends to have high coupling in another. Coupling trajectories will be displayed against time from lights-off and normalized night position. Sensitivity analyses will exclude windows containing transitions, arousals, apneas/hypopneas, or ectopic beats.

3.7 From a scalar trait to a physiological fingerprint

A single average may be stable even when spatial organization is not. We will therefore construct a fingerprint vector for each subject and night containing coupling by EEG channel and sleep stage. Fingerprints will be estimated with partial pooling so that channels or stages with fewer usable windows do not become spuriously extreme. Similarity between two fingerprints will be measured with Pearson correlation as the primary measure and cosine similarity as a sensitivity measure.

Fingerprint specificity will be evaluated by identification: for each repeat PSG, select the most similar baseline fingerprint from a candidate set containing the same person and other subjects recorded at the same site. Top-1 identification accuracy, rank of the true match, and the separation between within-person and between-person similarities will be compared with a subject-label permutation null distribution. Matching will be repeated after residualizing age, sex, stage composition, site, and signal quality.

3.8 Repeat-PSG analysis

For subjects with at least two technically adequate PSGs, visits will be separated into prespecified intervals (for example, less than 6 months, 6–24 months, and more than 24 months) when sample size permits. The visit-level scalar coupling estimate will be the empirical Bayes subject effect from a model standardized to a common distribution of sleep stages and covariates, rather than the raw nightly mean.

Long-term reliability will be assessed using a two-way mixed-effects absolute-agreement ICC for visit-level coupling, with subject and visit/session represented separately.[11–13] We will also report Pearson and Spearman correlations, mean change, Bland–Altman limits of agreement,[14] and reliability by inter-visit interval. A random-slope model with elapsed time will test gradual drift. Repeat studies affected by major treatment changes, acute illness, medication changes, or a different montage will be retained only in labeled sensitivity strata when such metadata are available.

Evidence that the fingerprint “comes back” requires convergence of three results: positive absolute-agreement reliability of the standardized scalar estimate, greater within-person than between-person fingerprint similarity, and above-chance blinded identification. Reliability that is confined to one site, montage, or uncorrected signal will not be called physiological stability.

Reliability depends on both aggregation and elapsed time (theoretical illustration)

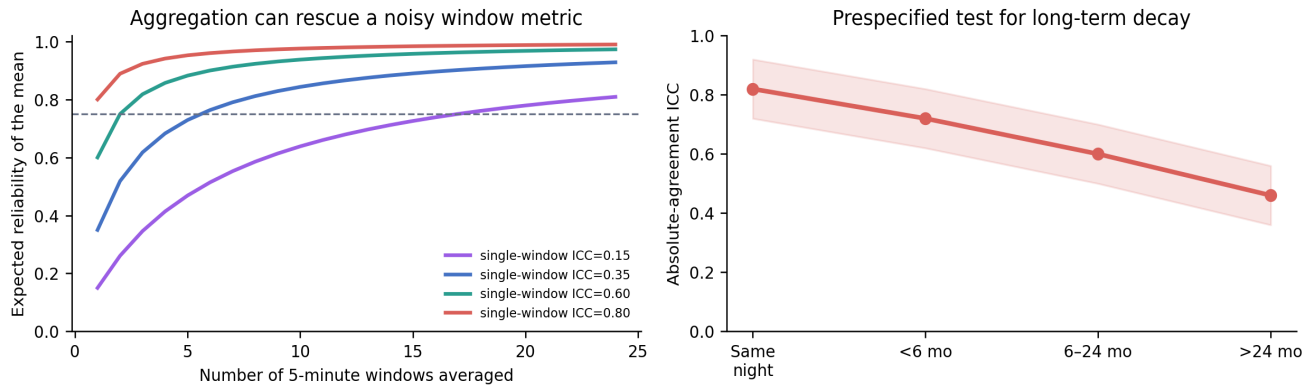


Figure 4. Reliability design. The Spearman–Brown relationship shows how aggregation across 5-minute windows can stabilize a noisy measure; the second panel depicts the prespecified test for decay with elapsed time. Curves and points are theoretical illustrations, not results.

The fingerprint must return in an independently acquired PSG

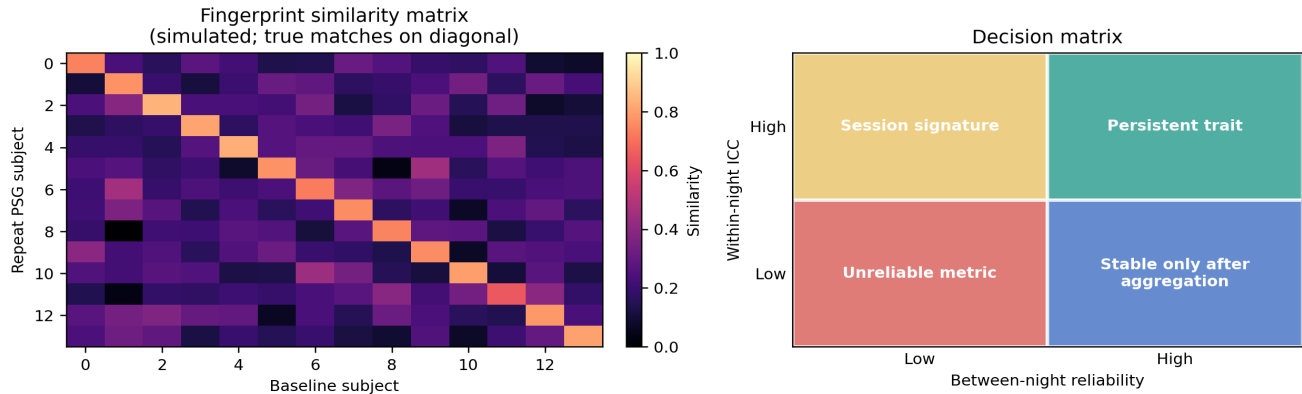


Figure 5. Repeat-PSG fingerprint test and interpretation. The similarity matrix illustrates blinded matching of follow-up to baseline fingerprints. The decision matrix prevents a high same-night ICC from being overinterpreted when between-night agreement is low. All displayed values are simulated.

3.9 Missingness and weighting

The mixed model naturally accommodates different numbers of usable windows under a missing-at-random assumption conditional on included covariates. No subject will receive greater scientific weight merely because their recording is longer: the likelihood model will account for repeated observations, and a sensitivity analysis will sample an equal number of windows per subject. We will summarize missing windows by subject, stage, cohort, and exclusion reason. Multiple imputation will be considered for missing subject-level covariates, but coupling outcomes will not be imputed.

3.10 Precision and sample size

The study will be powered for precision rather than a null-hypothesis test that ICC equals zero. Before final analysis, simulation will use the observed distribution of subjects and usable windows to estimate the expected width and coverage of the ICC confidence interval across plausible variance components and serial correlations. The repeat-PSG sample will be reported separately because long-term reliability depends more strongly on the number of repeated subjects than on the number of windows. Conclusions about trait persistence will be limited if the repeat sample cannot estimate reliability with useful precision.

3.11 Robustness and negative controls

Prespecified checks will include alternative window lengths (2.5, 10, and 20 minutes), equal-window sampling, robust mixed models, leave-one-cohort-out analysis, site-stratified estimates, alternative references, minimum-beat thresholds, and exclusion

of low-purity or transition windows. The principal negative controls are time-shifted R peaks, phase-randomized ECG, shuffled subject labels, non-heartbeat-locked EEG windows, and an ECG-to-EEG cardiac-field metric. These controls distinguish stable neurophysiology from stable anatomy, equipment, preprocessing, or leakage.

3.12 Multiple testing and reporting

The overall adjusted ICC is the single primary estimand. Stage-specific ICCs, channelwise effects, alternate metrics, and subgroup analyses are secondary and will use false-discovery-rate control within clearly defined families. We will publish point estimates and confidence intervals rather than dichotomize evidence at a single P-value. The analysis code, frozen configuration, exclusions, and model diagnostics will be retained with a reproducible environment.

4. Planned results presentation

Figures 1–5 in this Stage 1 manuscript define the workflow, estimands, confound structure, reliability analysis, and interpretation rules. They will remain in the final paper. After the frozen analysis is run, empirical panels will replace only the explicitly marked simulated panels; the decision rules and captions will be retained.

Empirical Figure E1: Coupling trajectories for representative subjects across the night, colored by sleep stage, with subject means and within-subject spread.

Empirical Figure E2: Variance decomposition showing $\sigma_{\text{subject}}^2$, σ_{within}^2 , and ICC with bootstrap confidence intervals before and after adjustment.

Empirical Figure E3: Stage-specific ICCs and the subject-by-stage correlation matrix.

Empirical Figure E4: Repeat-PSG reliability: baseline versus follow-up coupling, Bland–Altman plot, fingerprint-similarity distributions, and identification accuracy against permutation chance.

Table 1: Cohorts, subjects, repeat subjects, PSG duration, usable windows, channels, and inter-visit intervals.

Table 2: Unadjusted and adjusted variance components and ICCs.

Table 3: Sensitivity analyses and negative controls.

5. Interpretation framework

An ICC near zero would indicate that two windows from the same person are little more similar than windows from different people; coupling would be predominantly state-like or noisy at the chosen timescale. An intermediate ICC would support both trait and state contributions and motivate aggregation across multiple windows. A high adjusted ICC would indicate that subject identity explains a large share of residual coupling variation within a night.

The strength of the trait claim will depend on repeat PSGs. Four patterns are possible. High within-night and high between-night reliability would support a persistent physiological trait. High within-night but low between-night reliability would imply a session-specific state or recording signature. Low within-night but high reliability of nightly averages would indicate a weak momentary signal that becomes stable after aggregation. Low reliability at both levels would argue against using the metric as an individual fingerprint.

ICC is population- and context-dependent: it can increase because subjects are more heterogeneous or decrease because measurement noise is greater. Therefore, the manuscript will always present raw variance components, cohort composition, window reliability, and long-term agreement alongside ICC. “Trait” will denote reproducible person-level variation under the studied recording conditions, not immutability or causal biological identity.

6. Discussion

This study asks a simple but consequential question: is brain–heart coupling more variable across one person’s night or across people? Five-minute estimation exposes the temporal variation that nightly averaging conceals. Variance decomposition then expresses the answer on an interpretable scale, while repeat PSGs test whether the same individual signature survives independent acquisition months or years later.

The design explicitly separates state, trait, and artifact. Sleep-stage adjustment prevents sleep architecture from masquerading as identity. Serial-correlation modeling prevents adjacent windows from overstating precision. Cardiac-field and shuffled-event controls test whether the metric reflects neural coupling rather than electrical leakage. Site-matched fingerprint identification tests whether a classifier recognizes a person rather than a laboratory. Finally, absolute agreement across repeat visits demands more than rank-order similarity. This layered approach follows the broader reliability literature in neuroscience, where session, site, and measurement error can be decomposed rather than collapsed into one coefficient.[15]

The key outcome is not simply whether ICC is statistically nonzero. The useful scientific question is how much variance belongs to the subject, how much remains within the subject, how precisely both are estimated, and whether the person-level component returns in a new night. Together, these analyses provide a stringent test of brain–heart coupling as an individual physiological trait.

7. Open decisions to freeze before analysis

1. Exact primary coupling definition, EEG channels, frequency range, epoch, baseline, and post-heartbeat interval.
2. Cardiac-field-artifact correction method and the criterion for a correction failure.
3. Minimum accepted heartbeats and minimum usable windows per subject and stage.
4. Covariate availability and harmonization across cohorts.
5. The repeat-PSG interval bins and handling of treatment or montage changes.
6. Thresholds used for descriptive labels such as negligible, moderate, substantial, and high ICC.

These decisions must be timestamped before inspecting subject-level ICC or repeat-night matching results.

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